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1. Artificial Neural Networks and Perceptrons

An Artificial Neural Network (ANN) is defined as a computational model inspired by the biological nervous system, designed to recognize complex patterns across high-dimensional data. The fundamental processing unit is the Perceptron or Artificial Neuron, which calculates the weighted sum of inputs and passes this scalar through an activation function.

2. Activation Functions and Non-Linearity

Without non-linear activation functions, any deep multi-layer neural network collapses into a simple linear regression regardless of depth. ReLU (Rectified Linear Unit) is defined as f(x) = max(0, x), preventing gradient vanishing for positive activations. Sigmoid scales inputs between 0 and 1, frequently used in binary output layers. Softmax normalizes logit vectors into probability distributions summing to 1.0.

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3. Gradient Descent Optimization

Gradient Descent is defined as an iterative optimization algorithm used to minimize the objective loss function by updating parameter weights in the opposite direction of the gradient vector. The weight update rule is w_new = w_old - alpha * grad(L), where alpha denotes the learning rate parameter.

4. Backpropagation Calculus & Regularization

Backpropagation refers to the systematic application of the calculus Chain Rule to compute partial derivatives of the loss function with respect to every weight throughout all network layers. Overfitting occurs when a neural network memorizes training set noise rather than generalizing underlying structural features. Dropout randomly deactivates a fraction of neurons during training.